NON-WSQ COURSE • C34

Microsoft 365 Copilot for Excel

Version v3.0 · 11 October 2026

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

www.tertiarycourses.com.sg | enquiry@tertiaryinfotech.com | +65 6100 0613

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COURSE ADMINISTRATION

Welcome & Housekeeping

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About the Trainer



Dr. Alfred Ang

AI Trainer & Engineer

- PhD-qualified trainer specialising in AI, software engineering and developer tooling.
- Builds AI agents and Microsoft 365 Copilot workflows for business teams.
- Trains professionals across Singapore on practical, project-based AI skills.
- Certifications: PMP, AWS, Microsoft, CompTIA, CKA, CKAD.

linkedin.com/in/angchewhoe

github.com/alfredang

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ICE-BREAKER

Let's Know Each Other

1. Your name, organisation and role.
2. One HR, recruitment or finance report you rebuild in Excel every week or month.
3. The correction you find yourself giving Copilot (or a colleague) again and again.
4. Whether you can see the Copilot button in Excel, and on which app (Windows, Mac or web).

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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Access Course Materials

1. Open the C34 course page or the trainer-supplied Drive folder
2. Open the courseware links for this course
3. Download the Slides, Learner Guide and Lesson Plan
4. Open the labs folder for all 8 lab folders
5. Keep the files together as you work

Course page: tertiarycourses.com.sg/microsoft-365-copilot-for-excel.html

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SCHEDULE

Lesson Plan — 1 Day, 7.5 instructional hours

Course sequence

- Topic 1: Excel Fundamentals with Copilot (Labs 1–3)
- Topic 2: Skills and Rules (Labs 4–6)
- Topic 3: Interactive Dashboards with Copilot, Skills and Rules (Labs 7–8)

Day format

- Daily timing
  - 9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
- Instructional time
  - 7.5 hours · 8 connected labs
- Learning arc
  - Concept → demonstrate → build → verify → discuss → recap

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COURSE ROADMAP

Eight Labs in Three Topics

Excel Fundamentals

- Lab 1: Data Prep & Data Quality
- Lab 2: Formulas & XLOOKUP
- Lab 3: PivotTables, PivotCharts & Visuals

Skills & Rules

- Lab 4: Workbook Rules & Personalisation
- Lab 5: Built-in Skills: Visual & Finance
- Lab 6: Custom Skills: Calendar

Interactive Dashboard

- Lab 7: Interactive Dashboard
- Lab 8: Dashboard Skill, Refresh & Share

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Excel Fundamentals

Use Copilot in Excel's Chat only, Plan and Allow editing modes to prepare and clean data, write formulas and XLOOKUPs, and build PivotTables, PivotCharts and charts, verifying every result.

2

Skills & Rules

Make Copilot's work consistent and repeatable with personalisation, workbook rules (.Rules), built-in skills and custom skills that you create with Copilot.

3

Interactive Dashboard

Combine Copilot, rules and skills to build a highly visual, interactive and refreshable dashboard, and verify and share it responsibly.

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LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

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CORE CONCEPTS

Copilot in Excel for HR and Recruitment

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OVERVIEW

The BrightPath Scenario

1

BrightPath Talent Solutions

A fictional Singapore recruitment, flexible-staffing and HR-outsourcing firm.

2

Recruitment data

Candidate pipeline, job requisitions, monthly funnel metrics, interviews and client orders.

3

Business and workforce data

HR-outsourcing client finances, a headcount budget and internal workforce data.

4

Your role

HR / talent analyst who uses Copilot, rules and skills to turn these workbooks into verified dashboards.

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TOPIC 01

Excel Fundamentals with Copilot

Chat only, Plan and Allow editing · setup, data sources and the model picker · data prep and automated cleanup · formulas and XLOOKUP · PivotTables, PivotCharts and charts

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C34 OPERATING PRINCIPLE

Copilot does the work. Rules and skills make it repeatable. Your checks make it trustworthy.

Describe the task precisely, write the standards down once, and reconcile every number before it reaches a people decision.

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Ten Things Copilot in Excel Does Now

1. **Allow editing, Plan, Chat only**
   Three ways to work with Copilot - Lab 1.

2. **Analyse the data**
   Find patterns and answer questions - Labs 1, 7.

3. **Bring in other data**
   Attach another workbook as context - Lab 2 challenge.

4. **Model selector**
   Compare models on a hard question - Lab 1.

5. **Automated cleanup**
   Tidy messy data with Clean Data - Lab 4.

6. **Personalisation**
   Preferences that follow you - Lab 4.

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Ten Things Copilot in Excel Does Now (cont.)

7 Workbook rules
Standards stored in the file (.Rules) - Labs 4, 7, 8.

8 Write formulas
Describe the calculation you need - Lab 2.

9 Create new spreadsheets
Start from research or a blank workbook - Lab 3 challenge.

10 Skills
Make the work repeatable - Labs 5, 6, 8.

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Key Concepts — Excel Fundamentals with Copilot

1

Pick the right mode

Chat only answers without touching the sheet. Plan shows its steps first. Allow editing makes the change.

2

Keep or undo every change

After Copilot edits, select Done to keep it or Undo to reverse it. Version history is your safety net.

3

Turn your data into a table

Ask Copilot to format the range as a named table, e.g. tblApplications. Copilot gives better answers on tables.

4

Name the columns in your prompt

'Count hires by Source in tblApplications for 2026' beats 'analyse my data'.

5

Describe the formula in plain English

'Look up Recruiter_Name from tblRecruiters' - Copilot writes the XLOOKUP and explains it.

6

Check every number

Compare Copilot's answer with your own COUNTIFS, SUMIFS or PivotTable total.

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WHAT YOU'LL DO

Hands-On Labs — Excel Fundamentals with Copilot

Lab 1

- **Data Prep & Data Quality**

– Explore and Question Recruitment Pipeline Data

Lab 2

- **Formulas & XLOOKUP**

– Generate and Verify Formula Columns and XLOOKUPs for Job Requisitions

Lab 3

- **PivotTables, PivotCharts & Visuals**

– Build PivotTables, PivotCharts and Recruitment Charts with Copilot

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What Is Data Prep?

WHAT IT IS

- Turning a raw export into a clean table: one header row, one record per row, one kind of value per column.

WHY IT MATTERS

- Formulas, PivotTables, charts and Copilot only stay correct on a tidy table.

CHECK IT

- Can you say in one sentence what one row means?

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Try It in Excel — What Is Data Prep?

DO IT IN EXCEL

- Click in the data and press Ctrl+T (Format as Table).
- Table Design > Table Name: tblApplications.
- One header row, no blank rows, no merged cells.
- Grain: one row = one application.

THEN CHECK

- New rows join the table automatically.
- Every column holds one kind of value.

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Why Data Quality Matters

WHAT IT IS

- Good data is complete, consistent, unique and valid. Common problems: duplicates, spelling variants, text numbers.

WHY IT MATTERS

- Every count and chart inherits the errors: 'Linked In' and 'LinkedIn' become two sources.

CHECK IT

- Profile before you analyse: rows vs unique IDs, distinct values, data types.

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Try It in Excel — Why Data Quality Matters

CHECK IT IN EXCEL

- =ROWS(tblApplications) vs =COUNTA(UNIQUE(tblApplications[Application_ID]))
- =UNIQUE(tblApplications[Source]) lists every spelling.
- Or open the Source filter drop-down to see the variants.

THEN CHECK

- 242 rows vs 240 unique IDs = 2 duplicates.
- Fix in a helper column; keep the original.

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Chat Only, Plan and Allow Editing

WHAT IT IS

- Three modes: Chat only answers, Plan shows its steps first, Allow editing makes the change.

WHY IT MATTERS

- Copilot can edit your workbook. Picking the mode first keeps you in control.

CHECK IT

- After every edit, look at the highlighted changes, then select Done or Undo.

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Try It in Copilot — Chat Only, Plan and Allow Editing

TYPE THIS IN COPILOT

- Chat only: What does one row in tblApplications represent?
- Plan: Flag duplicate Application_IDs and standardise Source. Show me the steps first.
- Allow editing: Format this range as a table named tblApplications.

THEN CHECK

- Chat only left the sheet unchanged.
- Plan waited for your approval.
- Done keeps a change; Undo removes it.

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The Copilot Pane — What Each Control Does

1

Copilot button or floating icon

Home > Copilot, or the floating icon at the bottom right. Right-click the icon to dock it or move it to the ribbon.

2

Model picker (top)

Auto picks the model for each task. Or choose a Claude (Anthropic) or GPT (OpenAI) model for this session.

3

Mode (above the prompt box)

Chat only answers. Plan shows its steps and waits. Allow editing (the default) makes the change.

4

+ Add work content

Attach a file, choose plugins & skills, create workbook rules, or switch web and work sources.

5

Settings (...)

Personalization for your preferences; Manage plugins & skills to turn skills on or off.

6

New chat and history

Start a new chat when you switch task or change a setting; reopen earlier chats from history.

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Choosing a Model

1

Auto (recommended)

Copilot chooses the model for each task. Start here.

2

Claude - Anthropic

e.g. Claude Opus and Sonnet models. Deep reasoning models suit complex analysis.

3

GPT - OpenAI

OpenAI's GPT models.

4

Which one?

Try another model if an answer is weak. Premium models have fair-use limits. Always check the result.

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Setup, Data Sources and the Model Picker

WHAT IT IS

- Copilot needs an eligible licence and a current Excel. The + menu adds files and sources; the top menu picks the model.

WHY IT MATTERS

- Most 'Copilot is missing' problems are setup. Naming the right file beats letting Copilot guess.

CHECK IT

- Leave the model on Auto. When you compare models, check both against the same formula.

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Try It in Copilot — Setup, Data Sources and the Model Picker

SET IT UP

- Save as .xlsx in OneDrive; Calculation Options = Automatic.
- + > attach a file, or type / and the file name.
- Model picker (top): Auto, or a Claude or GPT model.
- New chat, then ask the same question to compare.

THEN CHECK

- The Copilot button is active, not grey.
- Both models agree with your COUNTIFS.

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Cleaning Data with Copilot

WHAT IT IS

- Ask Copilot to make the table, list the problems and fix them in a helper column. Data > Clean Data fixes spaces and capitals.

WHY IT MATTERS

- Copilot finds problems in seconds - but cleanup fixes typing, not meaning.

CHECK IT

- Count rows and unique IDs before and after. Flag problems before you fix them.

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Try It in Copilot — Cleaning Data with Copilot

TYPE THIS IN COPILOT

- Format this range as a table with headers and name it tblApplications.
- How many rows and how many unique Application_IDs are there?
- List the distinct values in Source and flag any spelling variants.

THEN CHECK

- 242 rows but 240 unique IDs = 2 duplicates found.
- No row deleted after the clean-up.

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Explore and Question Recruitment Pipeline Data

LAB 1

You are a talent analyst at BrightPath Talent Solutions, a fictional Singapore recruitment and staffing firm. You will meet Copilot's three modes on a real task: Allow editing turns the H1 2026 pipeline export into a named table, Chat only describes it without touching the sheet, and Plan shows you its steps before it fixes duplicate IDs and inconsistent Source labels. You will then answer pipeline questions, compare two AI models on the same question, and build a highlighted, sorted and filtered review view. Every number is checked with a formula.

You'll build

BrightPath_Candidate_Pipeline.xlsx in OneDrive with table tblApplications, a Source_Clean column, duplicate IDs highlighted, Hired rows highlighted, a verified source-by-hires answer and a filled Prompt_Log

1 Save to OneDrive	2 Create the table	3 Chat only: describe	4 Plan: fix quality	5 Ask, compare, filter

Tools: Excel for Microsoft 365 or Excel for the web · Copilot (Chat only, Plan, Allow editing, model picker) · BrightPath_Candidate_Pipeline.xlsx

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Sample Prompts — Lab 1

1. Describe: Describe tblApplications: what one row represents, each column's data type, the row count and the Applied_Date range. Do not change the workbook.

2. Data quality: List any Application_ID that appears more than once in tblApplications and any Source value that is not in the canonical list on About_This_Data.

3. Plan first: Plan how you would flag duplicate Application_IDs and standardise Source. Show me the steps before you change anything.

4. Clean helper column: Add a column Source_Clean that trims spaces from Source and replaces 'Linked In' with 'LinkedIn'. Do not change the original Source column.

5. Question: Using tblApplications, count applications and hires by Source_Clean and show the hire rate. Sort by hires, highest first.

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Check Your Results — Lab 1

1

Rows in tblApplications

242 rows, 240 unique Application_IDs

2

Duplicate IDs (orange)

APP-0018 and APP-0020 - 4 orange cells: 2 near the top, 2 in the last rows

3

Fixed Source cells (yellow)

5 cells: 3 'Linked In' + 2 'JobStreet '

4

LinkedIn count in Source_Clean

69

5

Hires

22 in total; LinkedIn has the most (8)

6

Technology + Interview filter

9 rows; all 242 back after clearing it

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What Is a Formula in Excel?

WHAT IT IS

- A formula starts with = and calculates from cells and functions. In a table, use names: =[@Filled]/[@Openings].

WHY IT MATTERS

- Formulas are auditable and recalculate when the data changes. A typed number does neither.

CHECK IT

- Click any result and read its formula in the formula bar.

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Try It in Excel — What Is a Formula in Excel?

TRY IN EXCEL

- Fill rate: =[@Filled]/[@Openings]
- Open or closed: =IF([@Close_Date]="","Open","Closed")
- Count: =COUNTIFS(tblRequisitions[Status],"Open")

THEN CHECK

- Typed once, it fills the whole column.
- No typed numbers in calculated columns.

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Table References: What Does @ Mean?

WHAT IT IS

- In a table, @ means 'this row'. [@Filled] is the Filled value in the same row as the formula.

WHY IT MATTERS

- One formula fills every row, reads like the rule, and keeps working when rows are added or sorted.

CHECK IT

- @ = this row. No @ = the whole column.

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Try It in Excel — Table References: What Does @ Mean?

READ IT LIKE THIS

- =[@Filled]/[@Openings] - this row's Filled / this row's Openings
- Same as =G3/F3 on row 3, but readable and safe to sort
- =SUM(tblRequisitions[Filled]) - the whole Filled column

THEN CHECK

- You only see @ inside an Excel table.
- =SUM([@Filled]) adds just one row - drop the @ for totals.

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Reading Excel References: !, :, $ and [@[...]]

WHAT IT IS

- ! = on that sheet. : = a range. $ = locked when copied. [@[Column]] = this row's value, same as [@Column].

WHY IT MATTERS

- Copilot's replies and formulas use them all - reading them shows which cells changed and where a formula looks.

CHECK IT

- Does the reference point at the right sheet, the right range and the right fixed input?

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Try It in Excel — Reading Excel References: !, :, $ and [@[...]]

READ IT LIKE THIS

- Requisitions!P2:P61 - cells P2 to P61 on the Requisitions sheet
- Control!B1 - cell B1 on the Control sheet, locked with $
- [@[Fee_Rate]] - this row's Fee_Rate, same as [@Fee_Rate]

THEN CHECK

- Press F4 while editing a reference to add or remove $.
- Sheet names with spaces need quotes: 'Dash Data'!B2.

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Reading a Formula: IF, AND and @

THE FORMULA, PIECE BY PIECE

- =IF(AND([@Status]="Open", [@[Days_Open]]>45), "Overdue", "")
- [@Status]="Open" - is this row's Status Open?
- [@[Days_Open]]>45 - has this row been open more than 45 days?
- AND(test1, test2) - TRUE only if both are TRUE
- IF(test, "Overdue", "") - show Overdue, otherwise leave blank

WHY THE @?

- @ = this row: each requisition is checked on its own row.
- Without @, [Status] means all 60 values at once - error or wrong result.
- [@Status] and [@[Days_Open]] mean the same kind of thing.
- Check: 13 rows show Overdue.

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What Is XLOOKUP?

WHAT IT IS

- XLOOKUP finds a key in one table and returns the matching value from another table.

WHY IT MATTERS

- Joins data by a shared ID instead of copy-paste - and updates when either table changes. Simpler and safer than VLOOKUP.

CHECK IT

- Count 'Not found' results: each one is a missing or mistyped key.

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Try It in Excel — What Is XLOOKUP?

THE SYNTAX

- =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])
- =XLOOKUP([@Recruiter_ID], tblRecruiters[Recruiter_ID], tblRecruiters[Recruiter_Name], "Not found")
- =COUNTIF(tblRequisitions[Recruiter_Name], "Not found")

THEN CHECK

- JOB-2002: R05 returns Siti Hamid.
- Keys must match exactly - watch extra spaces.

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Writing Formulas and XLOOKUPs with Copilot

WHAT IT IS

- Describe the calculation in plain English and Copilot writes the formula or XLOOKUP, and explains it.

WHY IT MATTERS

- A formula can run without errors and still be wrong, e.g. a monthly instead of an annual salary.

CHECK IT

- Work out one row by hand and compare the column total with your own formula.

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Try It in Copilot — Writing Formulas and XLOOKUPs with Copilot

TYPE THIS IN COPILOT

- Add Recruiter_Name by looking it up from tblRecruiters using Recruiter_ID.
- Add Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate, then a total row.
- Explain the formula in the selected cell.

THEN CHECK

- JOB-2002: 4 x S$4,900 x 12 x 25% = S$58,800.
- No 'Not found' in the lookup column.

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Generate and Verify Formula Columns and XLOOKUPs for Job Requisitions

LAB 2

BrightPath's permanent-recruitment desk tracks 60 client job orders. You will ask Copilot to add an XLOOKUP for the recruiter's name and four calculated columns written from business rules, ask it to explain each formula, hand-check one requisition and reconcile column totals to independent formulas on the Control sheet. You will then ask Copilot which recruiter generated the most estimated fees and verify the answer.

You'll build

BrightPath_Job_Requisitions.xlsx with Recruiter_Name, Days_Open, Fill_Rate, Est_Fee and Ageing_Flag columns in tblRequisitions, overdue requisitions highlighted, and a Control sheet that reconciles the totals

1 Lookup	2 Days open	3 Fill rate & fee	4 Ageing flag	5 Reconcile

Tools: Excel · Copilot (Allow editing, Chat only) · XLOOKUP · structured references · BrightPath_Job_Requisitions.xlsx

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Sample Prompts — Lab 2

1. Lookup: Add a column Recruiter_Name that looks up Recruiter_Name from tblRecruiters using Recruiter_ID with XLOOKUP. Return "Not found" if missing.

2. Duration: Add a column Days_Open: if Close_Date is blank use Control!B1 minus Open_Date, otherwise Close_Date minus Open_Date.

3. Ratio: Add a column Fill_Rate = Filled / Openings formatted as a percentage.

4. Business rule: Add a column Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate, formatted as S$ with no decimals.

5. Flag: Add a column Ageing_Flag that shows "Overdue" when Status is Open and Days_Open > 45, otherwise blank.

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Check Your Results — Lab 2

1 **Total estimated fees (Control B3 and B4)**
Both S$1,030,512

2 **Average days to fill (B5)**
34.6 days (29 filled requisitions)

3 **Overdue requisitions (B6 and B7)**
Both 13

4 **Sample row JOB-2002**
Est_Fee S$58,800, Days_Open 36

5 **Recruiter_Name lookup**
Every row has a name - no "Not found"

6 **Top recruiter by fees**
Aisha Rahman - S$221,760

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What Are PivotTables and PivotCharts?

WHAT IT IS

- A PivotTable summarises a table by dragging fields into Rows, Columns, Values and Filters. A PivotChart charts it.

WHY IT MATTERS

- Answers 'how many by X' in seconds, without formulas - and you can reshape it any time.

CHECK IT

- Is the value Sum, Count or Average? Did you Refresh after the data changed?

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Try It in Excel — What Are PivotTables and PivotCharts?

DO IT IN EXCEL

- Click in tblMonthly > Insert > PivotTable > New Worksheet.
- Rows: Month_Label · Values: Sum of Hires.
- PivotTable Analyze > PivotChart > Line.
- Data > Refresh All after the data changes.

THEN CHECK

- Grand total = SUM(tblMonthly[Hires]).
- The chart follows the PivotTable.

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PivotTables and PivotCharts from a Prompt

WHAT IT IS

- Describe the PivotTable and the PivotChart you want, and Copilot builds both, linked to your data.

WHY IT MATTERS

- The PivotTable total lets you check the chart, and both update when the data grows.

CHECK IT

- The PivotTable grand total equals a SUM or SUMIFS of the source column.

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Try It in Copilot — PivotTables and PivotCharts from a Prompt

TYPE THIS IN COPILOT

- Create a PivotTable from tblMonthly with Month_Label in rows and Sum of Hires.
- Create a line PivotChart of total hires by month from it, with data labels.
- Put both on a new sheet named Pivot_Trend.

THEN CHECK

- Grand total = SUM(tblMonthly[Hires]).
- The chart is a PivotChart, not a pasted picture.

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Choosing and Refining Charts

WHAT IT IS

- Pick the chart from the question, then refine it with one follow-up prompt per change.

WHY IT MATTERS

- The wrong chart type misleads. A finding-style title tells leaders what to see.

CHECK IT

- Could someone get the main message from the title in 5 seconds?

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Try It in Copilot — Choosing and Refining Charts

TYPE THIS IN COPILOT

- Create a bar chart of 2026 hires by Sector, sorted high to low.
- Change the title to: Technology led hiring in H1 2026.
- Put Cost_per_Hire on a secondary axis as a line, formatted as S$.

THEN CHECK

- Axes labelled with units.
- Values still match the PivotTable.

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Build PivotTables, PivotCharts and Recruitment Charts with Copilot

LAB 3

BrightPath's leadership wants to see how hiring is moving and which sectors and sourcing channels perform best. You will ask Copilot to build PivotTables and PivotCharts from 18 months of funnel metrics and a sourcing summary, choose each chart from the question it answers, refine titles, labels, sort order and axes with follow-up prompts, and reconcile every chart to an independent SUMIFS check.

You'll build

BrightPath_Recruitment_Metrics.xlsx with PivotTables and four refined charts on the Charts sheet: a monthly hires line PivotChart, a sorted 2026 hires-by-sector bar PivotChart, a 2026 recruitment funnel and a hire-rate vs cost-per-hire combo chart

1 Pivot + line PivotChart	2 Sector PivotChart	3 Funnel	4 Source combo	5 Refine & check

Tools: Excel · Copilot (Allow editing) · PivotTables, PivotCharts · line, bar, funnel and combo charts · BrightPath_Recruitment_Metrics.xlsx

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Sample Prompts — Lab 3

1. PivotTable: Create a PivotTable from tblMonthly with Month_Label in rows, Sector in columns and Sum of Hires as values.

2. PivotChart: Create a line PivotChart from the Pivot_Trend PivotTable showing total hires by month. Show data labels and gridlines.

3. Bar chart: Create a bar chart of 2026 hires by Sector, sorted from highest to lowest.

4. Funnel: Create a funnel chart of 2026 Applications, Screened, Interviews, Offers and Hires with data labels.

5. Combo chart: Create a combo chart of Hire_Rate as columns and Cost_per_Hire as a line on a secondary axis.

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Check Your Results — Lab 3

1 Total hires (Pivot_Trend)
1,065 = SUM(tblMonthly[Hires])

2 Peak month
2025-11 with 68 hires

3 2026 hires by sector
Technology first (84); Pivot total 361

4 Funnel
6,616 > 3,229 > 1,385 > 495 > 361

5 Best source
Referral: highest hire rate (8.7%); MyCareersFuture: lowest cost per hire (S$126)

6 Charts sheet
2 PivotCharts + funnel + combo in a 2 x 2 grid (A1:Q37), clean style, finding-style titles, labelled axes and alt text

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TOPIC RECAP

Recap - Excel Fundamentals with Copilot

1. LO1 · Modes: Chat only to ask, Plan to review the steps, Allow editing to change - then Done or Undo.

2. LO1 · Data prep: Named tables, duplicates and spelling variants flagged and fixed, row counts unchanged.

3. LO1 · Formulas: Formulas and XLOOKUPs described in plain English, explained, and hand-checked on one row.

4. LO1 · PivotTables and charts: PivotTables, PivotCharts and refined charts, each tied to a SUM or SUMIFS total.

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TOPIC 02

Skills and Rules

Personalisation · workbook rules on a .Rules sheet · built-in formatting, visualisation and finance skills · custom skills in SKILL.md · using Copilot to write a skill

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Key Concepts — Skills and Rules

1

Personalisation = you

Preferences that follow you to every workbook, e.g. 'Never merge cells'. Settings (...) > Personalization.

2

Rules = this workbook

A visible sheet named .Rules with one rule per cell in column A. The rules travel with the file.

3

Skills = a repeatable recipe

Saved step-by-step instructions you call by name with @, in any workbook.

4

Start with built-in skills

@chart-design for charts, @theme-design for styling, and finance skills such as @unit-economics and @check-model.

5

Let Copilot write your skill

Describe the task; Copilot writes the SKILL.md. Save it in its own OneDrive folder and select Refresh.

6

Still check the result

Rules and skills guide Copilot. They do not guarantee the numbers are right.

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WHAT YOU'LL DO

Hands-On Labs — Skills and Rules

Lab 4

- **Workbook Rules & Personalisation**

– Clean Messy Client Orders with Cleanup, Personalisation and Rules

Lab 5

- **Built-in Skills: Visual & Finance**

– Analyse HR-Outsourcing Finances with Built-in Skills

Lab 6

- **Custom Skills: Calendar**

– Use Copilot to Build a Custom Interview-Calendar Skill

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Personalisation: Preferences That Follow You

WHAT IT IS

- Your own preferences for Copilot, applied in every workbook. Settings (…) > Personalization.

WHY IT MATTERS

- Say a correction once instead of in every prompt. A prompt that asks for something else still wins.

CHECK IT

- Start a New chat, then check a new chart or table follows your preferences.

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Try It in Copilot — Personalisation: Preferences That Follow You

ADD THESE PREFERENCES

- Use column charts rather than bar charts unless labels are long.
- Use navy #1F4E79 and teal #2A9D8F as the main chart colours.
- Never merge cells.
- Give PivotTable fields readable names, e.g. Hires.

THEN CHECK

- Saved to your account, not the file.
- New chats follow them.

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Workbook Rules: The .Rules Sheet

WHAT IT IS

- A visible sheet named .Rules, one rule per cell in column A. Copilot follows it in this workbook.

WHY IT MATTERS

- Short prompts, consistent results - and anyone who opens the file gets the same rules.

CHECK IT

- Give a one-line prompt and check the result follows every rule.

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Try It in Copilot — Workbook Rules: The .Rules Sheet

TYPE THIS IN COPILOT

- + > Create workbook rules (adds the .Rules sheet).
- Status must be one of: Open, In Progress, On Hold, Completed, Cancelled. WIP = In Progress.
- Create Orders_Clean with a clean copy of Orders_Raw, following the workbook rules.

THEN CHECK

- Exactly 5 statuses and 5 regions.
- A hidden .Rules sheet is ignored.

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Clean Messy Client Orders with Cleanup, Personalisation and Rules

LAB 4

BrightPath's order system exports client orders that several people typed by hand: 'WIP', 'in progress' and 'Ongoing' all mean In Progress, regions are written 23 ways and client names carry 'Pte Ltd' suffixes. You will save your own preferences in Personalization, try Copilot's automated cleanup, then create a .Rules sheet that defines the workbook's standards. With the rules in place, a one-line prompt builds a clean table, and the same rules shape the XLOOKUP and budget columns you add next.

You'll build

Your Personalization (saved to your account) and BrightPath_Client_Orders_Raw.xlsx with a .Rules sheet, a clean tblOrdersClean table (5 statuses, 5 regions, 12 clients), Rate, Delivered_Value, Budget_Used and Over_Budget columns, and a Checks sheet that proves the clean-up

1 Personalise	2 Automated cleanup	3 Create .Rules	4 Apply the rules	5 Rules + XLOOKUP

Tools: Excel · Copilot (Allow editing, Chat only) · Personalization · Clean data with Copilot · .Rules worksheet · XLOOKUP

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Sample Prompts — Lab 4

1. Create rules: Create a .Rules worksheet based on the formatting already used in this workbook.
2. Add a rule: Add a rule to my .Rules worksheet that all charts should use navy #1F4E79 and teal #2A9D8F.
3. Apply rules: Create a new sheet Orders_Clean with a clean copy of Orders_Raw, following the workbook rules.
4. Find variants: List the distinct values in Status and Region with counts, and group the ones that mean the same thing.
5. Rules + lookup: Add Rate from tblRateCard by Owner and Delivered_Value = Hours_Logged x Rate to tblOrdersClean.

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Check Your Results — Lab 4

1

Orders and distinct values (Checks B1:B4)

64 orders; 5 statuses, 5 regions, 12 clients

2

Status counts

Open 10, In Progress 24, On Hold 6, Completed 21, Cancelled 3

3

Paused orders

ORD-302 and ORD-303 are On Hold

4

Total budget (B6)

S$3,250,500

5

Delivered value and over budget (B7, B8)

S$1,961,030; 10 orders over budget

6

Sample ORD-326

358 h x S$115 = S$41,170 vs budget S$36,500 - Over_Budget Yes

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Key Finance Measures: CAC, LTV, Churn and Payback

WHAT IT IS

- CAC is what it costs to win a client; LTV is what a client is worth; payback is how fast CAC is earned back.

WHY IT MATTERS

- Together they show whether growth pays off. A common benchmark: LTV:CAC of 3 or more.

CHECK IT

- Write the period and each definition before any skill calculates them.

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The Formulas — Key Finance Measures: CAC, LTV, Churn and Payback

THE FORMULAS

- CAC = sales & marketing spend / new clients
- ARPA = MRR / active clients
- LTV = ARPA x gross margin / monthly churn
- Payback = CAC / (ARPA x gross margin)

THEN CHECK

- H1 2026: CAC S$14,748, LTV about S$80,433.
- LTV:CAC 5.5; payback 8.9 months.

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Built-in Skills: Formatting, Visualisation and Finance

WHAT IT IS

- Ready-made skills you call with @: @chart-design, @theme-design, @brandkit and ~20 finance skills.

WHY IT MATTERS

- They package expert know-how - chart design, model audits, unit economics - in one line.

CHECK IT

- Hold the skill to your definitions and reproduce its headline numbers with formulas.

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Try It in Copilot — Built-in Skills: Formatting, Visualisation and Finance

TYPE THIS IN COPILOT

- @chart-design Create a combo chart of MRR and active clients by month.
- @unit-economics Analyse CAC, LTV and payback for H1 2026 using the definitions on About_This_Data.
- @check-model Audit Headcount_Budget. List issues; do not change anything yet.

THEN CHECK

- CAC on Checks = the skill's CAC.
- Budget total fixed: 3 errors found.

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Analyse HR-Outsourcing Finances with Built-in Skills

LAB 5

BrightPath's HR-outsourcing business sells monthly payroll and HR services to client companies. Finance wants to know whether winning clients pays off and whether next year's headcount budget is right. You will switch on skills in Manage plugins & skills, call them with @, and let each one do a specialist job: @chart-design builds the revenue chart, @unit-economics works out CAC, LTV and payback, @check-model finds the errors in the budget model, and @theme-design styles the workbook. A Checks sheet proves every headline number.

You'll build

BrightPath_HR_Outsourcing_Finance.xlsx with a skill-designed MRR chart, a Unit_Economics sheet, a corrected Headcount_Budget, a Checks sheet that agrees with the skills, and a consistent theme

1 Manage skills	2 @chart-design	3 @unit-economics	4 @check-model	5 @theme-design

Tools: Excel · Copilot (Allow editing, Plan, Chat only) · @chart-design · @unit-economics · @check-model · @theme-design

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Sample Prompts — Lab 5

1. Find skills: Which skills do you have for charts, formatting and finance? When would I use each one?
2. Chart skill: @chart-design Recommend the best chart for new vs churned clients by month, then create it.
3. Finance skill: @unit-economics Compare CAC and payback months by acquisition Channel for H1 2026.
4. Audit skill: @check-model Check this sheet for hardcoded numbers, broken formulas and totals that miss rows.
5. Style skill: @theme-design Apply a coordinated theme to this workbook.

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Check Your Results — Lab 5

1. **New clients and spend, H1 2026**
   38 clients; S$560,420

2. **CAC and ARPA**
   S$14,748; S$3,681

3. **Margin and monthly churn**
   44.9%; 2.06%

4. **LTV, LTV:CAC, payback**
   About S$80,433; 5.5; 8.9 months

5. **CAC by channel**
   Lowest Referral S$5,180; highest Events S$33,445

6. **@check-model fixes**
   E3, E6, E10: total S$1,437,934 -> S$1,703,052

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Custom Skills: The SKILL.md Recipe

WHAT IT IS

- A SKILL.md file in its own OneDrive folder: name and description at the top, plain-English steps below.

WHY IT MATTERS

- Turns a long, repeated prompt into a one-line @mention that works in any workbook.

CHECK IT

- Folder name = name: line. Select Refresh after every change.

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Try It in Copilot — Custom Skills: The SKILL.md Recipe

SET IT UP

- … > Manage plugins & skills > Custom skills > Create OneDrive folder.
- New folder: interview-calendar; upload SKILL.md into it.
- Custom skills > Refresh; turn the skill on.
- @interview-calendar Week starting 6 July 2026.

THEN CHECK

- The skill appears when you type @.
- It runs from one line.

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Using Copilot to Write and Improve a Skill

WHAT IT IS

- Describe the task in Chat only and Copilot writes the whole SKILL.md for you.

WHY IT MATTERS

- No coding needed - and a skill gets better each time you add a correction to it.

CHECK IT

- Read every step before saving, and test the skill on a copy first.

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Try It in Copilot — Using Copilot to Write and Improve a Skill

TYPE THIS IN COPILOT

- Write a SKILL.md named interview-calendar that turns tblInterviews into a weekly calendar and flags clashes.
- Return only the SKILL.md text, starting with the --- front matter.
- Update the skill so the summary shows hours per Interviewer.

THEN CHECK

- Week 1: 23 interviews, 3 clashes.
- Week 2 needs only one short prompt.

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Use Copilot to Build a Custom Interview-Calendar Skill

LAB 6

Every Friday a BrightPath coordinator turns next week's interview bookings into a calendar and hunts for double-bookings. That is a repeatable recipe, so it should be a skill. You will open the sample calendar skill in your OneDrive skills folder, ask Copilot to write a new SKILL.md for interview-calendar, save it in its own folder, refresh the skills list and call it with @. The same one-line prompt then builds the next week's calendar.

You'll build

An interview-calendar skill in your OneDrive skills folder, and BrightPath_Interview_Schedule.xlsx with Cal_2026-07-06 and Cal_2026-07-13 calendar sheets, each with clashes flagged and a summary

1 Skills folder	2 Copilot writes SKILL.md	3 Save & refresh	4 @interview-calendar	5 Reuse & improve

Tools: Excel · Copilot (Chat only, Allow editing) · Manage plugins & skills · OneDrive skills folder · SKILL.md

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Sample Prompts — Lab 6

1. Write a skill: Write a SKILL.md for a skill named weekly-recruiter-report that summarises interviews by Interviewer and Client for a given week, with a small bar chart.

2. Explain a skill: Read this SKILL.md and tell me what Copilot will do, step by step, and anything that is unclear.

3. Call a skill: @interview-calendar Build the calendar for the week starting Monday 6 July 2026.

4. Reuse: @interview-calendar Week starting 13 July 2026.

5. Clash check: List every interview clash in tblInterviews for 6-10 July 2026 with both Interview_IDs.

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Check Your Results — Lab 6

1

Week of 6 July

23 interviews = the COUNTIFS check

2

By day

Mon 7, Tue 5, Wed 5, Thu 3, Fri 3

3

Clashes flagged

INT-010/INT-011 Kevin Ong; INT-015/INT-016 Orchid Room; INT-018/INT-019 Aisha Rahman

4

Busiest interviewer

Aisha Rahman (9 interviews)

5

Week of 13 July

18 interviews, no clashes

6

Skill

interview-calendar appears when you type @ and runs from a one-line prompt

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TOPIC RECAP

Recap - Skills and Rules

1. LO2 · Personalisation: Your own preferences, saved once in Settings and applied in every workbook.

2. LO2 · .Rules: One workbook's standards in a visible .Rules sheet, so a one-line prompt cleans the data.

3. LO2 · Built-in skills: @chart-design, @unit-economics, @check-model and @theme-design - checked against formulas.

4. LO2 · Custom skills: A SKILL.md written with Copilot, saved in OneDrive and reused with one @mention.

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TOPIC 03

Interactive Dashboards with Copilot, Skills and Rules

Designing a highly visual dashboard · KPI definitions and style in .Rules · plan, build and polish with skills · slicers, timelines and refresh · a reusable dashboard skill · sharing people dashboards responsibly

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Key Concepts — Interactive Dashboards with Copilot, Skills and Rules

1

Define each KPI first

Write the formula in .Rules, e.g. Cost per hire = total sourcing cost / hires.

2

Put the standards in .Rules

KPI definitions, layout and colours in the rules keep the build prompt short.

3

Plan, then build

Plan mode shows the dashboard plan. Fix a wrong KPI before Copilot copies it into five charts.

4

Make it visual

KPI cards on top, charts in a grid, traffic-light colours, then @chart-design and @theme-design to polish.

5

Make it interactive

Slicers and a timeline filter every PivotTable and chart at once, and the KPI cards must follow.

6

Make it repeatable and safe

Save the recipe as a skill, test refresh with new rows, and hide groups under 5 people.

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WHAT YOU'LL DO

Hands-On Labs — Interactive Dashboards with Copilot, Skills and Rules

Lab 7

- **Interactive Dashboard**
  - Build an Interactive Recruitment Dashboard with Rules and Skills

Lab 8

- **Dashboard Skill, Refresh & Share**
  - Turn the Dashboard into a Reusable Skill, Refresh It and Share It Safely

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Designing a Highly Visual Dashboard

WHAT IT IS

- One page: KPI cards on top, charts in a grid, slicers at the side, colour that signals status.

WHY IT MATTERS

- Leaders read it in seconds - if every KPI has one agreed definition.

CHECK IT

- Every KPI has a formula, a period, a target and an owner.

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Try It in Copilot — Designing a Highly Visual Dashboard

TYPE THIS IN COPILOT

- Suggest 5 KPIs for a recruitment dashboard using the columns in tblApps.
- Write Offer acceptance rate = accepted / (accepted + declined), excluding Pending.
- Suggest a layout and a 3-colour palette for a one-page dashboard.

THEN CHECK

- 4-6 headline KPIs, not 20.
- Definitions and colours go into .Rules.

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Plan and Build with Rules and Skills

WHAT IT IS

- Rules hold the definitions and style; Plan mode shows the build; skills polish the result.

WHY IT MATTERS

- Fix a wrong KPI in the plan before it is copied into five charts.

CHECK IT

- Click each KPI card: it should hold a formula, not a typed number.

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Try It in Copilot — Plan and Build with Rules and Skills

TYPE THIS IN COPILOT

- Plan mode: Build a one-page H1 2026 recruitment dashboard following the workbook rules.
- @chart-design Review every chart: finding-style titles and the rule colours.
- @theme-design Apply a modern theme that keeps the rule colours.

THEN CHECK

- The plan uses the rules' definitions.
- Each card matches the Checks sheet.

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Interactivity: Slicers, Timelines and KPI Cards

WHAT IT IS

- Slicers and a timeline filter every PivotTable, PivotChart and KPI card on the page at once.

WHY IT MATTERS

- Leaders answer their own 'what about…?' questions - if the page is wired correctly.

CHECK IT

- Test one slicer value and one date range against COUNTIFS, then clear them.

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Try It in Copilot — Interactivity: Slicers, Timelines and KPI Cards

TYPE THIS IN COPILOT

- Add slicers for Sector, Source and Recruiter and a timeline on Hire_Date, connected to every PivotTable.
- Link the KPI cards to the PivotTables with GETPIVOTDATA.
- Which objects did not change when I selected Technology?

THEN CHECK

- Technology: cards = COUNTIFS checks.
- Apr-Jun 2026 hires = COUNTIFS on Hire_Date.

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Build an Interactive Recruitment Dashboard with Rules and Skills

LAB 7

BrightPath's Head of Recruitment wants one page that shows H1 2026 applications, hires, offer acceptance, time to hire, cost per hire and hires against target, and can be filtered by sector, source, recruiter and date. You will put the KPI definitions, layout and colours in a .Rules sheet, so the build prompt stays short. Copilot plans the dashboard in Plan mode, builds it, then @chart-design and @theme-design make it look professional. Finally you prove that every KPI, slicer and the timeline give the right numbers.

You'll build

BrightPath_Recruitment_Dashboard_Data.xlsx with a .Rules sheet, a Checks sheet, PivotTables on Dash_Data and a Dashboard sheet with six formula-driven KPI cards (one with traffic-light colours), four PivotCharts, Sector, Source and Recruiter slicers and a Hire_Date timeline

1 Write .Rules	2 Build checks	3 Plan mode	4 Skills make it visual	5 Slice & verify

Tools: Excel · Copilot (Plan, Allow editing, Chat only) · .Rules · @chart-design · @theme-design · PivotCharts, slicers, timeline

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Sample Prompts — Lab 7

1. Plan: Plan a recruitment dashboard following the workbook rules. Show me the plan first.

2. KPI card: Create a KPI card for Cost per hire linked to the PivotTable with GETPIVOTDATA, formatted as S$.

3. Chart skill: @chart-design Turn the hires-by-month chart into columns for hires and a line for Hire_Target.

4. Timeline: Add a timeline on Hire_Date and connect it to every PivotTable on Dash_Data.

5. Visual cue: Add data bars to the hires-by-recruiter table and traffic-light colours to the target card.

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Check Your Results — Lab 7

1. **Applications and hires**
   600 applications, 41 hires (no slicer selected)

2. **Offer acceptance rate**
   78.8% = 41 of 52; 10 pending excluded

3. **Time to hire, cost per hire**
   27.8 days; S$348.20

4. **Hires vs target (amber)**
   93.2% = 41 of 44

5. **Slicer: Sector = Technology**
   124 applications, 5 hires, 42.0 days

6. **Timeline: Apr-Jun 2026**
   22 hires; no KPI card holds a typed number

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Workforce Measures: Headcount, Attrition and Tenure

WHAT IT IS

- Headcount = active staff on a date. Attrition = exits / average headcount. Tenure = years since hire.

WHY IT MATTERS

- Small definition changes move the rate. Written in .Rules, every rebuild uses the same ones.

CHECK IT

- Average headcount = (opening + closing) / 2 - not closing only.

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The Formulas — Workforce Measures: Headcount, Attrition and Tenure

THE FORMULAS

- Headcount: =COUNTIFS(tblEmployees[Status],"Active")
- Average headcount = (opening + closing) / 2
- Attrition = exits in period / average headcount

THEN CHECK

- 29 / 223.5 = 13.0% attrition.
- No group under 5 people shown alone.

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A Reusable Dashboard Skill and the Refresh Test

WHAT IT IS

- Save the dashboard recipe as a skill, then prove the page updates when new rows arrive.

WHY IT MATTERS

- Next month's dashboard takes one line - and a refresh test catches static pictures and typed numbers.

CHECK IT

- Add test rows, refresh all, and confirm each KPI moves by the expected amount.

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Try It in Copilot — A Reusable Dashboard Skill and the Refresh Test

TYPE THIS IN COPILOT

- Write a SKILL.md named hr-workforce-dashboard that reads .Rules first and checks every KPI.
- @hr-workforce-dashboard Build the dashboard for the 12 months ending 2026-06-30.
- Append the 8 new joiners to tblEmployees and refresh all PivotTables.

THEN CHECK

- Active headcount 237 -> 245.
- Every chart updated.

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Sharing People Dashboards Responsibly

WHAT IT IS

- HR dashboards hold sensitive data. Share totals and averages, never individual records.

WHY IT MATTERS

- Anyone who can open the file sees everything in it, including hidden sheets.

CHECK IT

- No slicer combination shows a group of fewer than 5 people.

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Try It in Copilot — Sharing People Dashboards Responsibly

TYPE THIS IN COPILOT

- Which slicer combinations show fewer than 5 employees?
- List every element on this sheet that could identify a person.
- Write a 5-line leadership summary using only numbers on the dashboard.

THEN CHECK

- Privacy rule written in .Rules.
- Every number in the summary checked.

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Turn the Dashboard into a Reusable Skill, Refresh It and Share It Safely

LAB 8

BrightPath's HR Director needs the same kind of dashboard every month. Instead of repeating a long brief, you will ask Copilot to write an hr-workforce-dashboard skill, save it to OneDrive and run it on the workforce workbook, whose .Rules sheet holds the period, the attrition definition and the privacy rule. Then you will append eight July joiners and refresh to prove the dashboard is live, check that no slicer combination exposes fewer than 5 people, and ask Copilot for a short leadership summary.

You'll build

An hr-workforce-dashboard skill in OneDrive, and BrightPath_Workforce_Headcount.xlsx with a .Rules sheet, helper columns, a Checks sheet and an HR_Dashboard with six KPI cards, four charts and Location and Job_Level slicers - refresh-tested and privacy-checked

1 Write .Rules	2 Copilot writes the skill	3 @hr-workforce-dashboard	4 Refresh test	5 Privacy & summary

Tools: Excel · Copilot (Chat only, Plan, Allow editing) · .Rules · custom skill · PivotTables, slicers, refresh

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Sample Prompts — Lab 8

1. Write a skill: Write a SKILL.md named hr-workforce-dashboard that reads the .Rules sheet first and builds a verified workforce dashboard.

2. Run a skill: @hr-workforce-dashboard Build the workforce dashboard for the 12 months ending 2026-06-30.

3. Refresh: I added rows to tblEmployees. Refresh all PivotTables and tell me which KPI values changed.

4. Privacy: Which slicer combinations show a group of fewer than 5 employees? Suggest how to suppress them.

5. Explain: Explain how the Attrition rate card is calculated, step by step.

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Check Your Results — Lab 8

1. **Active headcount**
   237 before the refresh

2. **Exits and attrition**
   29 exits (25 voluntary); 13.0% attrition

3. **Hires and tenure**
   56 hires; average tenure about 3.1 years

4. **Highest-attrition department**
   Recruitment: 21.8% (12 exits)

5. **Refresh test**
   After adding the July joiners: headcount 245, 428 rows

6. **Privacy**
   No Employee_ID, salary, individual record or group under 5 on the dashboard

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TOPIC RECAP

Recap - Interactive Dashboards with Copilot, Skills and Rules

1. LO3 · Design: KPI definitions, layout and colours written in .Rules before anything is built.
2. LO3 · Build: Plan mode, then @chart-design and @theme-design; every KPI card matches the Checks sheet.
3. LO3 · Interact: Slicers, a timeline and GETPIVOTDATA cards tested against COUNTIFS.
4. LO3 · Reuse and share: A dashboard skill, a passing refresh test, and no group under 5 people on view.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Excel Fundamentals

Use Copilot in Excel's Chat only, Plan and Allow editing modes to prepare and clean data, write formulas and XLOOKUPs, and build PivotTables, PivotCharts and charts, verifying every result.

2

Skills & Rules

Make Copilot's work consistent and repeatable with personalisation, workbook rules (.Rules), built-in skills and custom skills that you create with Copilot.

3

Interactive Dashboard

Combine Copilot, rules and skills to build a highly visual, interactive and refreshable dashboard, and verify and share it responsibly.

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NEXT STEPS

Continue the Verified Workflow

1. Add your three most common corrections to Personalization.
2. Write a .Rules sheet for one workbook you use every month.
3. Turn one repeated report into a custom skill - let Copilot write the first draft.
4. Keep people decisions with accountable people: Copilot informs, it does not decide.

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Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Keep your downloaded C34 course files and use the lab checkpoints to restart safely.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.

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Thank You!

You can now use Copilot in Excel for the fundamentals, make its work repeatable with rules and skills, and build interactive dashboards - while keeping evidence, privacy and human judgement in control.

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